



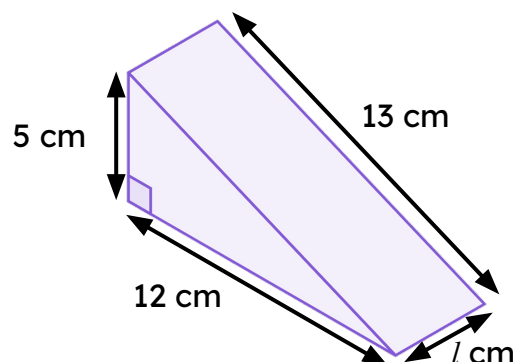
Problem solving with perimeter, area and volume

Task A: Surface area and volume

1) The surface area of this triangular prism is 240 cm^2 .

a) What is the length of the prism?

$$\begin{aligned} 2\left(\frac{1}{2} \times 5 \times 12\right) + 12l + 5l + 13l &= 240 \\ 60 + 30l &= 240 \\ 30l &= 180 \\ l &= 6 \end{aligned}$$



b) Hence calculate the volume of the prism.

$$\text{Volume} = \left(\frac{1}{2} \times 5 \times 12\right) \times 6 = 180 \text{ cm}^3$$

2) The volume of a cylinder is $324\pi \text{ cm}^3$ and its height is 9 cm.

Work out the cylinder's surface area.

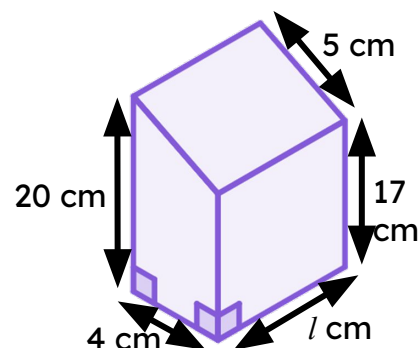
$$\begin{aligned} \text{Volume} &= 324\pi \\ \pi r^2 \times 9 &= 324\pi \\ \pi r^2 &= 36\pi \\ r^2 &= 36 \\ r &= 6 \end{aligned}$$

$$\begin{aligned} \text{Surface area} &= 2\pi r^2 + 2\pi rh \\ \text{Surface area} &= 2 \times 36\pi + 2\pi \times 6 \times 9 \\ \text{Surface area} &= 72\pi + 108\pi \\ \text{Surface area} &= 180\pi \text{ cm}^2 \end{aligned}$$

3) The surface area of this prism is 516 cm^2 .

Work out the volume of the prism.

$$\begin{aligned} \text{Surface area} &= 516 \\ 2\left(\frac{1}{2} \times (20 + 17) \times 4\right) + 20l + 4l + 17l + 5l &= 516 \\ 148 + 46l &= 516 \\ 46l &= 368 \\ l &= 8 \end{aligned}$$



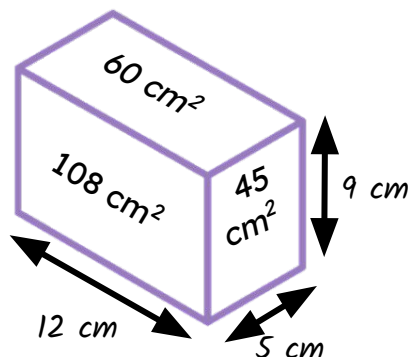
$$\text{Volume} = \left(\frac{1}{2} \times (20 + 17) \times 4\right) \times 8$$

$$\text{Volume} = 592 \text{ cm}^3$$



- 4) The areas of three faces have been given. The edges are integer amounts of centimetres.

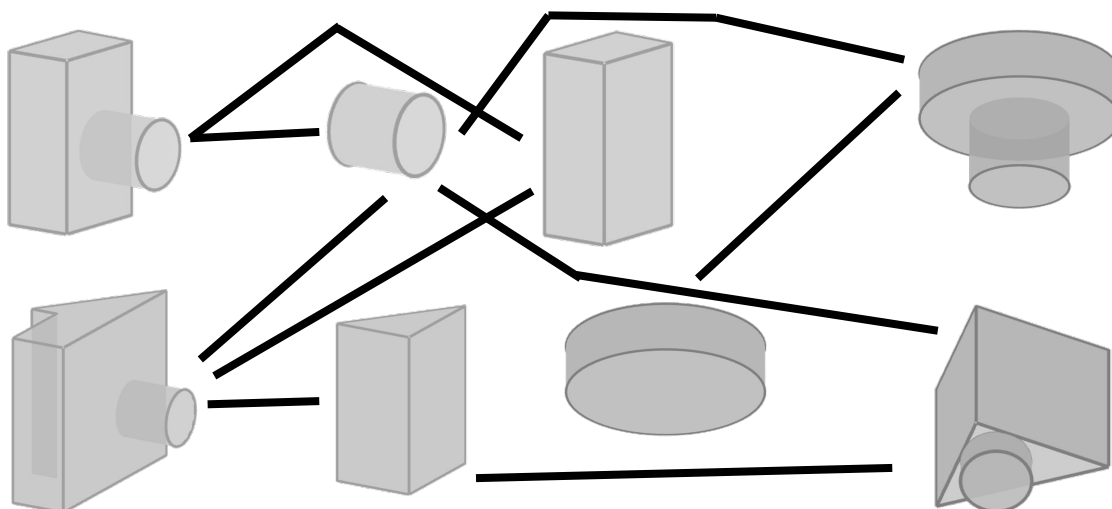
What is the volume of this cuboid?



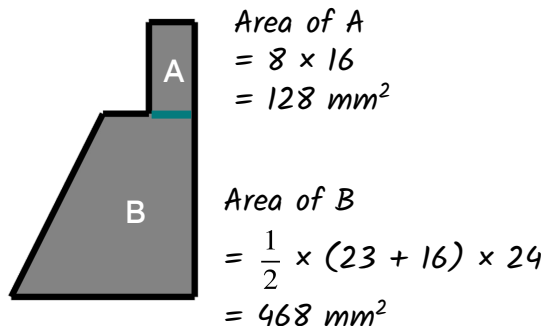
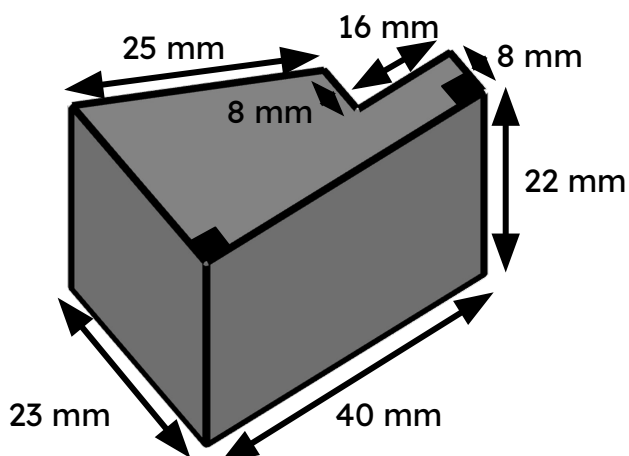
$$\text{Volume} = 12 \times 5 \times 9 = 540 \text{ cm}^3$$

Task B: Composite solids

- 1) Match the composite solid with its parts.



- 2) Calculate the volume of this prism.



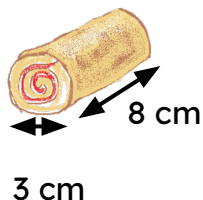
Area of cross-section
 $= 128 + 468 = 596 \text{ mm}^2$

Volume
 $= 596 \times 22 = 13\,112 \text{ mm}^3$



Task C: Problems in context

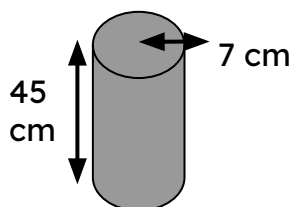
- 1) A baker makes 24 mini swiss rolls and wants to cover them in chocolate. How many square centimetres of chocolate will the baker have used?



$$\text{Surface area} = 2 \times \pi \times 1.5^2 + \pi \times 3 \times 8 = 28.5\pi \text{ cm}^2$$

$$\text{Chocolate used} = 24 \times 28.5\pi = 684\pi \text{ cm}^2$$

- 2) A metal pole is melted down and remoulded into cubes with edges of 5 cm. How many cubes can be made from this pole?



$$\text{Volume of pole} = \pi \times 7^2 \times 45 = 2205\pi \text{ cm}^3$$

$$\text{Volume of cube} = 5 \times 5 \times 5 = 125 \text{ cm}^3$$

$$\text{Amount of cubes} = 2205\pi \div 125 = 55.417\dots$$

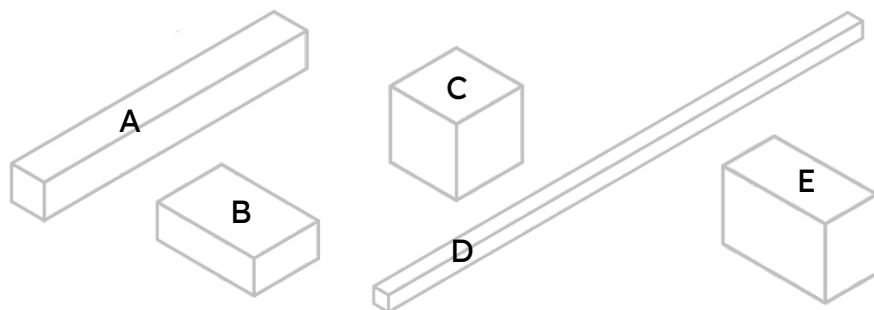
55 cubes can be made

- 3) One factor that affects the rate of dissolving is the contact between the solute and the solvent.

Here are some sugar lumps (solute) of the same mass. When put in water (solvent), which would have:

a) the most contact? D - 2008 mm²

b) the least contact? C - 600 mm²



	<i>l</i>	<i>w</i>	<i>h</i>
A	40	5	5
B	20	10	5
C	10	10	10
D	250	2	2
E	12.5	8	10

l = length (mm)
w = width (mm)
h = height (mm)

